

Module 1 – Worksheet 1

Multiple Choice Questions

Q.No 1. What is the first step in the problem-solving process?

- a) Implementing a solution
- b) Identifying the problem
- c) Analyzing the solution
- d) Evaluating alternatives

Answer:

Q.No 2. Which of the following is a key component of effective problem-solving?

- a) Ignoring possible solutions
- b) Jumping to conclusions
- c) Open-mindedness
- d) Avoiding collaboration

Answer:

Q.No 3. What does the acronym SMART stand for in the context of setting goals during problem-solving?

- a) Simple, Manageable, Achievable, Relevant, Timely
- b) Specific, Measurable, Achievable, Relevant, Time-bound
- c) Strategic, Meaningful, Attainable, Realistic, Timeless
- d) Structured, Motivating, Actionable, Reflective, Tangible

Answer:

Q.No 4. During the problem-solving process, what does brainstorming primarily involve?

- a) Criticizing ideas
- b) Generating ideas
- c) Implementing solutions
- d) Ignoring alternatives

Answer:

Q.No 5. Which of the following is a common barrier to effective problem-solving?

- a) Collaboration
- b) Lack of creativity
- c) Open communication
- d) Flexibility

Answer:

Q.No 6. What is the purpose of conducting a root cause analysis in problem-solving?

- a) Implementing quick fixes
- b) Identifying the underlying issues
- c) Ignoring the problem
- d) Avoiding responsibility

Answer:

Q.No 7. In the problem-solving context, what does the term "SWOT" stand for?

- a) Specific, Workable, Organized, Timely
- b) Strengths, Weaknesses, Opportunities, Threats
- c) Systematic, Wise, Organized, Thoughtful
- d) Solving, Weighing, Organizing, Tracking

Answer:

Q.No 8. What is the significance of prototyping in problem-solving?

- a) Ignoring feedback
- b) Testing and refining solutions
- c) Avoiding alternative solutions
- d) Rushing into implementation

Answer:

Q.No 9. What role does critical thinking play in problem-solving?

- a) Encouraging unquestioning acceptance
- b) Discouraging analysis
- c) Facilitating logical reasoning and evaluation
- d) Avoiding complexity

Answer:

Q.No 10. What is a common pitfall to avoid when solving problems in a group setting?

- a) Encouraging collaboration
- b) Groupthink
- c) Embracing diversity of thought
- d) Open communication

Answer:

Module 1 – Worksheet 2

Fill in the blanks

Q.No.1. In top-down design, the problem is broken down into _____ subproblems.

Answer:

Q.No.2. The top-level module in top-down design represents the _____ of the problem.

Answer:

Q.No.3. The process of breaking down a complex problem into manageable parts in top-down design is often referred to as _____.

Answer:

Q.No.4. Top-down design promotes _____ by allowing different teams to work on specific modules independently.

Answer:

Q.No.5. The top-level module in top-down design is sometimes called the _____ module.

Answer:

Match the following

PART A		PART B		ANSWER
(1)	Divide and Conquer	(A)	An algorithmic technique that involves trying out different possibilities until a solution is found or all possibilities have been exhausted.	
(2)	Greedy Algorithms	(B)	A straightforward approach to problem-solving that involves systematically trying all possible solutions until the correct one is found.	
(3)	Backtracking	(C)	A problem-solving technique that involves breaking down a problem into smaller, more manageable subproblems to solve them independently and then combine the solutions.	
(4)	Dynamic Programming	(D)	A problem-solving strategy that makes locally optimal choices at each stage with the hope of finding a global optimum	
(5)	Brute force	(E)	A method for solving complex problems by breaking them down into simpler overlapping subproblems, and solving each subproblem only once, storing the solutions to avoid redundant computations.	
Answer (1) – (2) – (3) – (4) – (5) –				

Module 1 – Worksheet 3

Q.No 1 : Write the Algorithm for finding the greatest common divisor of two numbers

Hint : Euclidean Algorithm

a = 48 and b = 18 , then $\text{GCD}(a,b) = 6$

Solution

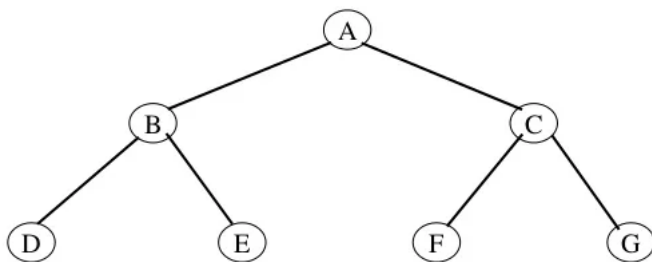
Q.No 2: Write an algorithm for predicting house prices based on features like size, location, and number of bedrooms

Hint :

1. Collect and Prepare Data
2. Define Features and Target
3. Initialize Model Parameters
4. Define the Hypothesis Function
5. Define the Cost Function – Mean Squared Error
6. Gradient Descent – minimizing the cost function
7. Training the Model
8. Make Predictions
9. Evaluate the Model
10. Deploy the Model

Solution:

Q. No 3 : Write the algorithm for Breadth First Search



Hint :

- a. Traverse Breadth wise
- b. Root Node : A

c. Output: A B C D E F G

Solution:

Q. No: 4. Calculate the Space Complexity of the following code

```
def linear_search(arr, target):  
    for element in arr:  
        if element == target:  
            return True  
    return False
```

Hint : The algorithm uses a constant amount of space

Solution :

Q. No: 5. Calculate the Space Complexity of the following code

```
def fibonacci(n):  
    fib = [0] * (n+1)  
    fib[0] = 0  
    fib[1] = 1  
  
    for i in range(2, n+1):  
        fib[i] = fib[i-1] + fib[i-2]  
  
    return fib[n]
```

Hint : The algorithm uses a constant amount of space

Solution :

Q. No: 6. Calculate the Time Complexity of the following code

```
def binary_search(arr, target):  
    low, high = 0, len(arr) - 1  
  
    while low <= high:  
        mid = (low + high) // 2  
        if arr[mid] == target:  
            return mid  
        elif arr[mid] < target:  
            low = mid + 1  
        else:  
            high = mid - 1  
    return -1
```

Hint : Binary search repeatedly halves the search space until the target is found or the search space becomes empty

Solution :

Q. No: 7. Calculate the Time Complexity of the following code

```
def quick_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    pivot = arr[len(arr)//2]  
    left = [x for x in arr if x < pivot]  
    middle = [x for x in arr if x == pivot]  
    right = [x for x in arr if x > pivot]  
  
    return quick_sort(left) + middle + quick_sort(right)
```


Hint : Worst Case - When the pivot selection consistently results in unbalanced partitions.

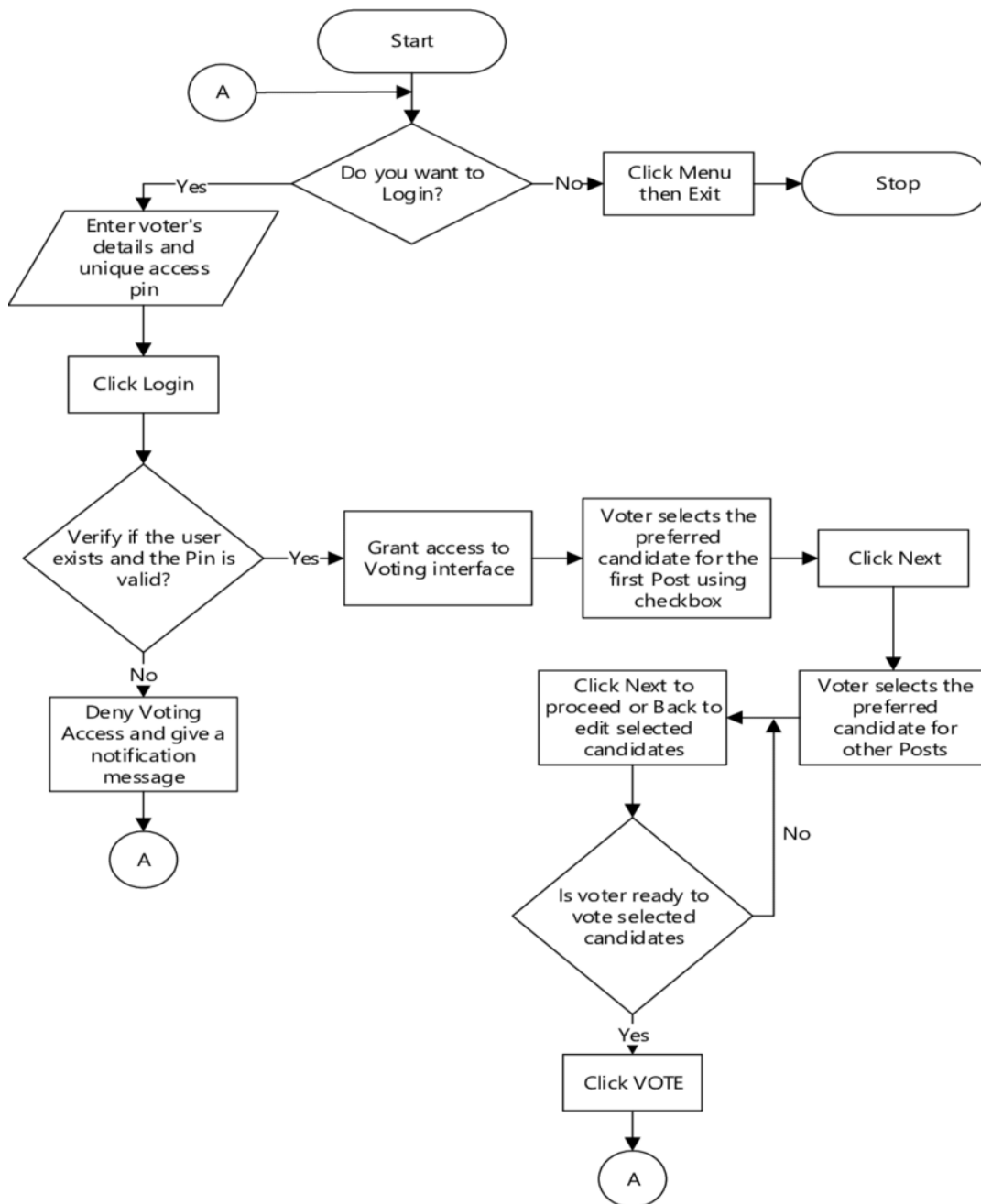
Solution :

Q. No 8. Draw the Flow chart for the Bubble sort Algorithm

Hint: The logic is to repeatedly go through the list, compare adjacent elements, and swap them if they are in the wrong order.

Solution

Q. No 9. Write the Pseudocode for the below flowchart



Hint: Voting System

Solution:



Q. No 10. Consider the following two algorithms, Algorithm A and Algorithm B, for finding the maximum element in an array. Analyze the time complexity of each algorithm and compare their efficiency.

Algorithm A: FindMaxA(arr)

Input: An array 'arr' of n elements

Output: The maximum element in the array

1. Set maxElement to the first element of the array (maxElement = arr[0])

Algorithm B: FindMaxB(arr)

Input: An array 'arr' of n elements

Output: The maximum element in the array

1. Set maxElement to the first element of the array (maxElement = arr[0])

2. For i from 1 to n-1: a. If arr[i] is greater than maxElement: i. Set maxElement to arr[i] 3. Return maxElement	2. For i from 1 to n-1: a. For j from i+1 to n: i. If arr[j] is greater than maxElement: - Set maxElement to arr[j] 3. Return maxElement
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Hint:
The time complexity of Algorithm A is $O(n)$, where 'n' is the size of the input array.
The time complexity of Algorithm B is $O(n^2)$, where 'n' is the size of the input array.

Solution